blood

722.CLINICAL ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION | NOVEMBER 5, 2020

Gut Microbiota-Induced Regulatory T Cells in Patients with Hematological Malignancies Receiving Allogeneic Hematopoietic Stem Cell Transplantation: Towards Deciphering a Role for These Tregs in aGVHD

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Prognosis for hematological malignancies treated with allogeneic stem cell transplantation (allo-HSCT) remains dismal and can often induce potentially deadly complications, such as graft *versus* host disease (GvHD). Steroids remain the first-line treatment for aGvHD. Nevertheless, new treatments are needed to improve allo-HSCT outcomes and reduce steroid complications. In this context, increasing evidences suggest that gut microbiota composition and the activity of regulatory T cells (Tregs) are involved in GvHD prevention (Elias and Rudensky, 2019, Br. J. Haematol. ; Shono and van den Brink, 2018, Nat. Rev. Cancer).

We have identified a novel FoxP3-negative IL-10-secreting Treg subset, named DP8α, enriched in the colon mucosa and present in blood, characterized by its $CD3^{+}/CD4^{+}/CD8\alpha^{LOW}/ CCR6^{+}/CXCR6^{+}$ phenotype and its TCR-specificity for the gut *Clostridium IV* bacterium *Faecalibacterium prausnitzii* (Sarrabayrouse *et al.*, 2014, PLoS Biol ; Godefroy *et al.*, 2018, Gastroenterology). Sizable fractions of these cells also expressed the membrane-bound ectonucleotidases CD39 and CD73 (gating strategy shown in *Figure 1*), which are directly involved in their suppressive activity *in vitro* (Godefroy *et al.*, 2018, Gastroenterology). *In vivo*, these molecules play key regulatory roles both by degrading extracellular ATP and thus limiting its inflammasome-related proinflammatory role, and by inducing the production of immunosuppressive adenosine. Accordingly, an alteration in adenosine/purinergic signaling has been suggested in GvHD occurrence (Deaglio *et al.*, 2007, J. Exp. Med. ; Wang *et al.*, 2013, PLoS ONE). Thus, we postulated that *F. prausnitzii*-reactive DP8α Tregs, whose suppressive activity depends on purinergic signaling, could bridge microbiota dysbiosis and GvHD incidence in allo-HSCT patients. In support of this, decreased levels of *Clostridium* bacteria, especially *Faecalibacterium spp,* in patient's stools, have been associated with aGvHD risk (Noor *et al.*, 2019, J Innate Immun). Moreover, mice gavaged with butyrate-producing and Treg-inducing *Clostridia* displayed milder GvHD and improved survival (Mathewson *et al.*, 2016, Nat. Immunol.).

Therefore, we recently analyzed 63 patients with hematological malignancies, who received allo-HSCT, among which 21 developed an aGvHD (pathologies and treatments are listed *Table I*). We quantified circulating DP8α Tregs and analyzed their expression of CD39 and CD73 in donors, before and after mobilization, and in patients at one week before and 1, 2, and 3 months post allo-HSCT. Strikingly, results clearly showed that aGvHD development was strongly associated with a lack of expression of CD73 by DP8α Tregs ($p<.0001$) (*Fig.2A*), but not by any other T cell subsets analyzed. DP8α frequency and their CD39 expression were similar in GvHD+ *versus* GvHD- patients. Importantly, CD73 decrease did not result from their corticotherapy since it was equally observed before and after treatment (*Fig.2B*) and since *in vitro* treatment of PBMCs with methylprednisolone did not alter CD73 expression whatsoever. Therefore, the CD73 decrease observed in aGvHD patients, appears relevant and likely related to the aGvHD condition itself. It is noteworthy that this cohort was rather heterogeneous in terms of hematological diseases, conditioning regimens and prophylaxis treatments (*Table I*). Nevertheless, this drastic CD73 decrease in aGvHD patients was observed across all these conditions, strongly strengthening its relevancy.

Altogether, these data strongly suggest that a CD73-dependent functional alteration of DP8α Tregs are, at least in part, involved in aGvHD occurrence. These results could therefore be used to predict GvHD risk and also give rise to the development of therapeutic strategies. Such innovative treatments could be based on the infusion of DP8α Tregs with appropriate features, since we have shown both the uniquely high proliferating potential and the stable immunosuppressive function of these cells *in vitro*. Moreover, administration of DP8α target antigens (*F. prausnitzii*-derived), in the form of peptides, proteins, or even bacterial lysates, such as probiotics, could also be foreseen to either directly *in vivo* expand these Tregs or indirectly through the induction of tolerogenic dendritic cells (Alameddine *et al*., 2019, Front Immunol) and subsequent Treg differentiation/priming to limit GvHD-related inflammation.

Figure 1

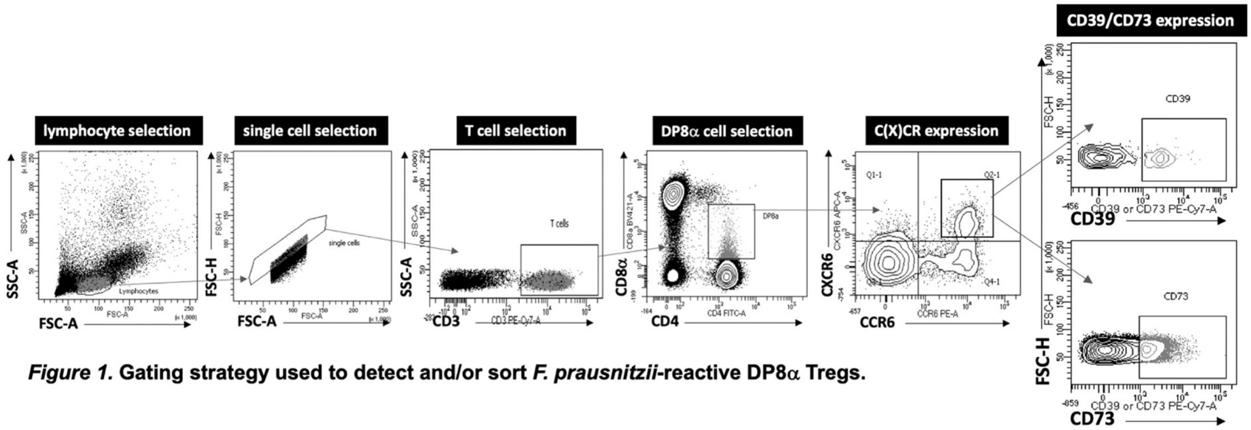

Figure 1. **Gating strategy used to detect and/or sort *F. prausnitzii*-reactive DP8α Tregs.**

		Healthy Donors (n=42)	Patients (n=63)
Gender	Female: n (%)	**20** (47,6%)	**28** (44,4%)
Age	Years: Median (min, max)	**54** (36, 72)	**60** (22, 71)
Diseases	AML ALL MDS Lymphoma Myelofibrosis Others	N/A	**24** (38,1%) **8** (12,7%) **14** (22,2%) **9** (14,3%) **6** (9,5%) **2** (3,2%)
Types of Donors	Haplo-identical	N/A	**27** (42,9%)
	Matched 10/10 : - siblings - unrelated	N/A	**10** (15,9%) **26** (41,2%)
Conditioning	MAC	N/A	**9** (14,3%)
	RIC	N/A	**54** (85,7%)
GvHD Prophylaxis	Ciclosporin	N/A	**2** (3,2%)
	Ciclosporin / ATG	N/A	**3** (4,8%)
	Ciclosporin / MMF / ATG	N/A	**21** (33,3%)
	Ciclosporin / Methotrexate	N/A	**1** (1.6%)
	PTCy	N/A	**7** (11,1%)
	PTCy / Ciclosporin / MMF	N/A	**19** (30,2%)
	PTCy / Ciclosporin / MMF / ATG	N/A	**10** (15,9%)
Complications	no GvHD	N/A	**27** (42,9%)
	Grade II aGvHD Grade III, IV aGvHD cGvHD Deaths by GvHD	N/A	**18** (28,6%) **3** (4,8%) **5** (7,9%) **4** (6,3%)
	Relapses Deaths by Relapse	N/A	**13** (20,6%) **5** (7,9%)
	Other Deaths	N/A	**6** (9,5%)
Organs targeted by aGvHD	Skin	N/A	**10** (45%)
	Intestine	N/A	**9** (41%)

Table I.
Clinical characteristics of patients with hematological malignancies treated with allo-HSCT and of healthy donors. AML: acute myeloid leukemia, ALL: acute lymphoid leukemia, MDS: myelodysplastic syndrome, MAC: myeloablative conditioning, RIC: reduced intensity conditioning, PTCy: post-transplant cyclophosphamide, MMF: mycophenolate mofetil, ATG: antithymocyte globulin, aGvHD: acute GvHD, cGvHD: chronic GvHD, N/A: not applicable.

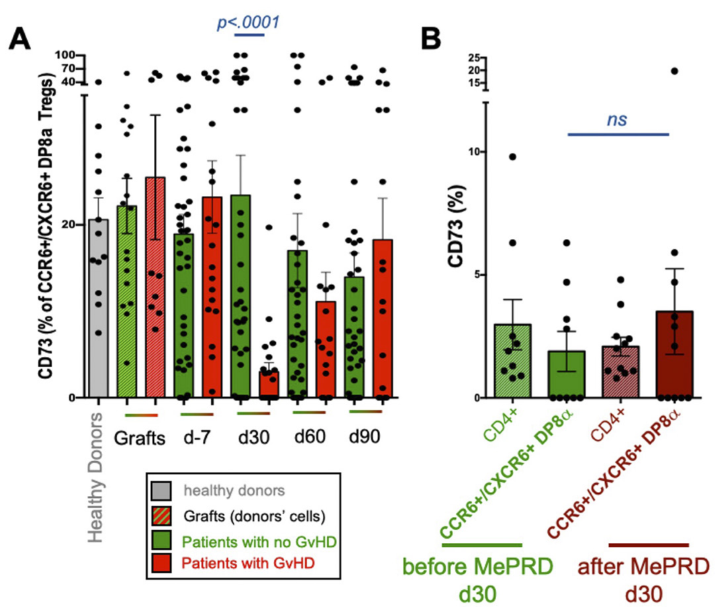

Figure 2. **Role of DP8α Tregs in GvHD development.** ***A.*** CD73 expression by blood CCR6+/CXCR6+ DP8α Tregs in patients receiving allo-HSCT developing GvHD (n=21) or not (n=42). CD73 expression by DP8α Tregs is also represented for healthy donors and donors' cells from the actual received grafts. ***B.*** CD73 expression by indicated cell subsets at d30 in GvHD+ patients before or after corticotherapy (MePRD, methyl prednisolone). Mann-Whitney test, p<.05 considered stastistically significant.

Disclosures

Chevallier: *Incyte Corporation:* Honoraria.

Author notes

* Asterisk with author names denotes non-ASH members.